Subathira Rajavel

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OBJECTIVE

To Contribute to academia as an Assistant Professor by combining excellence in teaching, research, and service, fostering critical thinking among students, and addressing real-world challenges through scholarly work.

EDUCATION

SSN College of Engineering, Chennai, India 2025
M.E. Medical Electronics – CGPA: 9.2 (First Class with Distinction)

Dr. NGP Institute of Technology, Coimbatore, India 2023
B.E. Biomedical Engineering – CGPA: 8.69 (First Class with Distinction)

Velammal Matric Hr. Sec School, Madurai, India 2019
Higher Secondary Certificate (HSC) – 72%

SPJ Matric Hr. Sec School, Madurai, India 2017
Secondary School Leaving Certificate (SSLC) – 90%

EXPERIENCE

Indian Institute of Technology Madras Jan 2026 – Mar 2026
Project Associate - Biotech Department Chennai, India

- Developed **3+ interactive serious games** for neurodiverse children, improving cognitive, motor, and imitation skills
- Designed imitation-based learning modules for children with Autism Spectrum Disorder (ASD), enhancing engagement levels by **30%** during trials
- Developed and deployed games: **Dunk-it!!, Froot Shoot, Dart Knight** as part of a collaborative team
- Integrated real-time motion tracking using **MediaPipe**, improving interaction accuracy by **25%**
- Successfully launched applications on **Microsoft Store** and pilot tested in **2+ special education schools** (Clarke School, Mirra School, Chennai)

SRM Easwari Engineering College Sep 2025 – Dec 2025
Junior Research Fellow Chennai, India

- Worked on **DHR-funded Sign Language Recognition Project**
- Developed Indian Sign Language recognition system using **MediaPipe + Deep Learning**, achieving **high classification accuracy**
- Created and curated a structured dataset with **1000+ gesture samples** for model training
- Enhanced and completed a partially developed system, improving performance efficiency by **20%**
- Contributed to **patent registration** of the developed system
- Designed laboratory layout utilizing project funding and coordinated with **multiple vendors** for setup

SSN College of Engineering Mar 2024 – Jun 2025
Research Associate Chennai, India

- Worked on XR-based surgery simulation project in collaboration with University of Bologna, Italy
- Developed VR-based surgical training systems using Unity and Blender with **real-time interaction capabilities**
- Focused on immersive healthcare technologies and haptic-based interaction systems, improving simulation realism
- Designed and implemented 3D anatomical models and surgical instruments for procedure simulation
- Contributed to development of haptic feedback integration and interaction logic for realistic user experience

Excel College of Engineering May 2025 – Aug 2025
Assistant Professor – Biomedical Engineering Namakkal, India

- Handled lectures for **3+ core subjects** including Biomedical Instrumentation, Bio-informatics, and VR in Healthcare
- Mentored **60+ students** in laboratory sessions and guided academic projects
- Managed Instrumentation Laboratory and assisted Bio-signal Processing Lab sessions
- Contributed to curriculum development and academic mentoring

PROJECTS

VR-Based Surgery Simulator with Haptic Feedback

- Developed a Total Knee Replacement (TKR) simulator using Unity and Blender with realistic 3D human knee models and surgical tools
- Implemented real-time interaction, incision simulation, and complete surgical workflow modeling
- Integrated **Touch Haptic Device** for force feedback, depth-based incision resistance, and tactile sensation
- Simulated soft tissue deformation and bone interaction to enhance procedural realism
- Published work in an **IEEE Conference**
- Drafted journal paper under international collaboration (Italy) and initiated patent registration

Sign Language Recognition System

- Developed real-time Indian Sign Language recognition system using MediaPipe and deep learning models
- Performed hand landmark extraction, gesture classification, and model training
- Created structured dataset improving recognition accuracy and model performance
- Contributed to patent filing under DHR-funded project
- Journal paper completed and submitted for publication

Sleep Apnea Detection from ECG Signals

- Developed SVM-based classification model using MATLAB for apnea detection
- Performed ECG signal preprocessing, feature extraction, and classification

Tissue Waste Disposer - Working Prototype

- Designed an automated biomedical tissue waste incineration system for safe disposal
- Focused on healthcare safety standards and environmental compliance
- Presented at international conference and received **Best Project Recognition Award**

PUBLICATION

- Divya B, Kavitha A, Pravin Kumar S, Subathira R, *"Comparative Analysis of Realistic 3D Models for VR-based Surgery Simulation"*, 3rd International Conference on Artificial Intelligence and Machine Learning Applications, April 2025
- Sign Language Recognition System using MediaPipe and Deep Learning – Journal paper submitted
- VR-Based Surgery Simulator with Haptic Feedback – Journal paper drafted and sent to Italy collaborators, patent under process

TECHNICAL EXPERTISE

Programming
C, C#, Python, MATLAB

Software & Development
Unity (VR Development), OpenCV, LabVIEW

Medical Imaging Tools
3D Slicer, ImageJ

Hardware
Meta Quest 3, Touch Haptic Device, Microsoft HoloLens

Designing Tools
Blender, EasyEDA

Documentation Tools
Microsoft Word, Excel, PowerPoint, LaTeX

INTERNSHIP & TRAINING

Perfint Private Limited May 2024
Internship – Research and Development & Quality Assurance

- Gained exposure to research and development processes in medical device systems
- Understood quality assurance procedures and regulatory standards in healthcare technology

National Institute of Technical Teachers Training and Research Mar 2024
Internship – 3D Asset Development in Blender and VR Applications using Unreal Engine

- Developed 3D models and explored game mechanics, supporting M.E. thesis development
- Gained hands-on experience in Blender and Unreal Engine for VR application development

Vel Multi-specialty Hospital, Palani Feb 2024
Clinical Immersion

- Observed emergency unit workflows, including response time of doctors and nurses, and identified operational challenges
- Proposed engineering-based solutions to improve efficiency and workflow
- Learned about emergency medical equipment and their manufacturers

Meenakshi Mission Hospital 2023
Clinical Training

- Worked with biomedical engineers to understand equipment maintenance and emergency repair procedures
- Gained knowledge in equipment procurement, maintenance, quality assurance, and calibration documentation

Deepam Academy of Medical Sciences May 2022
30-Day Online Training

- Completed training on diagnostic, treatment, operative, and post-operative medical equipment

Kovai Medical Center and Hospital 2022 – 2023
Hospital Training Phase 1 and Phase 2

- Studied hospital departments and working principles of medical devices across units
- Performed hands-on calibration training on infusion pumps and syringe pumps
- Gained knowledge of electrical safety and patient-device interaction in medical equipment
- Learned radiation safety guidelines and working principles of radiological equipment

Atheena Pandian Private Limited Mar 2021
In-Plant Training (3 Days)

- Received hands-on training in biomedical equipment servicing and maintenance
- Worked on servicing X-ray machines and oxygen generators

INDUSTRIAL VISITS

- **Leepra Technologies, Mysore (2022)** – Observed PCB assembly and system-level manufacturing processes
- **Labland Biotech Private Limited, Mysore (2022)** – Studied plant biotechnology applications and industrial practices

CERTIFICATIONS & ONLINE COURSES

- **NPTEL** – Biomedical Nanotechnology (Dec 2020, 65%)
- **NPTEL** – Biomechanics (Nov 2021, 60%)
- **NPTEL** – Biophotonics (Apr 2022, 60%)
- **ISSRT** – Osteoporosis and DXA Scanning (Sep 2021)
- **ISSRT** – Medical Imaging Equipment Procurement and Specifications (Aug 2021)

PROFESSIONAL STRENGTHS

- Interdisciplinary research focus integrating Biomedical Engineering, Artificial Intelligence, and Virtual Reality
- Hands-on experience in developing real-time healthcare applications and simulation systems
- Experience working in collaborative research environments with contributions to publications and patents
- Quick learner with the ability to adapt and apply emerging technologies to real-world medical challenges

HOBBIES

- Drawing and Creative Sketching
- Watching Movies and Exploring Visual Storytelling
- Reading Novels and Research Articles